KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (MECHANICAL ENGINEERING)

EMM 108: WORKSHOP PROCESSES AND PRACTICE I

2ND SEMESTER 2020/2021 **TIME: 2 HOURS**

INSTRUCTIONS:

1. This paper contains five (5) questions.
2. Question One (1) carries thirty (30) marks and the remaining four questions each carries twenty (20) marks.
3. Attempt Question One (1) and any other two questions from the remaining four questions.

QUESTION 1 (COMPULSORY)

a) Briefly discuss the causes of industrial accidents under the following:-
- Human Causes *(3 marks)*
- Environmental Causes *(3 marks)*
- Mechanical Causes *(3 marks)*

b) Describe the steps (showing the major operations) involved in making the dust scoop shown in Figure 1. Identify the appropriate tools for each operation. Also, state any five safety precautions to observe/care for the tools employed. *(12 marks)*

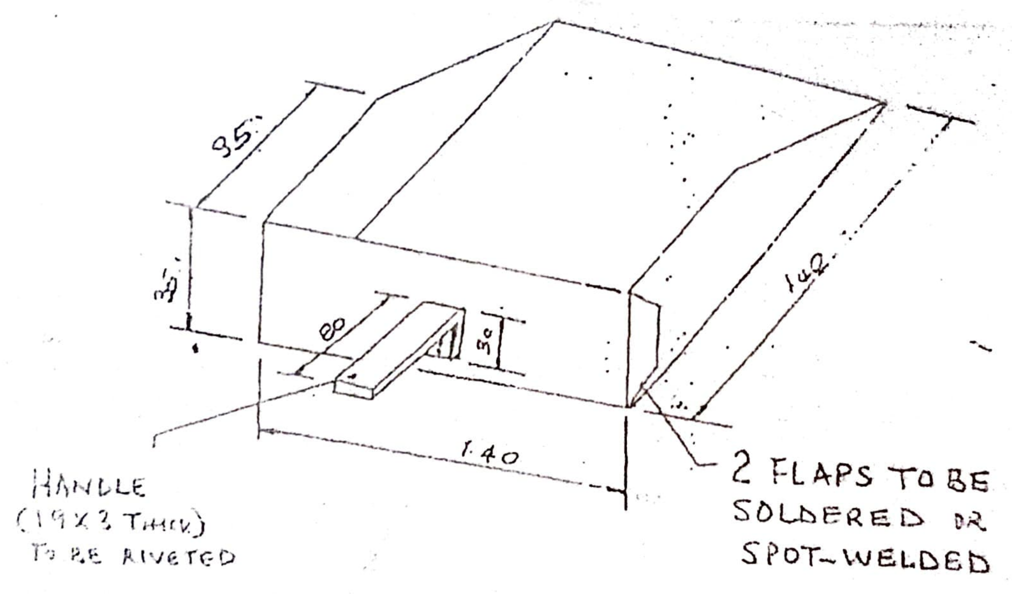

Figure 1

Notes:

- All dimensions in mm
- Edge and body flaps to be at 30°
- Drawing not to scale

c) A M105 metric system set of gauge blocks (slip gauges) contains blocks as shown in the table below

Range (in mm)	Steps (in mm)	Number of pieces
1.01 – 1.49	0.01	49
0.50 – 24.50	0.5	49
25 – 100	25	4
1.0025	-	1
1.005	-	1
1.0075	-	1

Explain how a length (stack) of 69.7825 mm can be built up using slip gauges without protector blocks and when using protector blocks of 1.5 mm thickness placed at the bottom and top of the stack. Use no more than four blocks. *(6 marks)*

d) A wedge shaped job was measured using a 250 mm sine bar and the top found to be level when gauge blocks totaling 46.376 mm were placed under one end of the rollers. What is the angle of the wedge? *(2 marks)*

e) Briefly state the function of a reamer *(1 mark)*

QUESTION 2

a) Differentiate between measuring instruments and gauges *(2 marks)*

b) With the aid of neat diagrams, briefly discuss the following gauges:-
- Ring gauges *(2 marks)*
- Plug gauges *(2 marks)*
- Snap gauges *(2 marks)*

c) Describe two conditions for sine bar to measure accurately *(2 marks)*

d) Distinguish the use of taps and dies in a fitting shop. *(2 marks)*

e) A combination set consist of a number of parts. Identify the parts and discuss their purposes. *(8 marks)*

QUESTION 3

a) Briefly define pinning as applied to files. *(1 mark)*

b) Identify the three file classification criteria and specify two examples for each classification. *(6 marks)*

c) Briefly write short notes on the following tools:-

- Dot and center punches *(2 marks)*
- Inside and outside calipers *(2 marks)*
- V-block *(1 mark)*
- Trammel *(1 mark)*

d) Briefly discuss the operation of sawing. *(4 marks)*

e) State two examples under each of the following tools:-

- Holding tools *(1 marks)*
- Striking tools *(1 mark)*
- Tightening tools *(1 mark)*

QUESTION 4

a) State four fundamental differences between soldering and brazing. *(4 marks)*

b) Briefly describe the sequential steps to follow during soldering. *(8 marks)*

c) What is a flux? State any three functions of a flux in brazing and soldering techniques. *(4 marks)*

d) State two properties of fluxes used in soldering. *(2 marks)*

QUESTION 5

a) Illustrate using sketches how the following readings may be set and worked out:-

- 13.44 mm of a depth micrometer having an accuracy of 0.01 mm
- 22.32 mm of a Vernier calipers having an accuracy of 0.02 mm *(4 marks)*

b) State the function of the following measuring instruments:-

- Telescopic gauge
- Screw thread micrometer *(2 marks)*

c) Describe with the aid of a diagram the working of an outside micrometer screw gauge. *(8 marks)*

e) With the aid of neat sketches, describe the process of wringing of slip gauges. *(6 marks)*